ASSIGNMENT

MongoDB_Queries

```
use("college");
```

Insert Multiple Student Records :

```
db.students.insertMany([
   {
      name : "sanskruti",
      age : 20,
      branch : "IT",
      marks : 98,
      city : "Nashik"
   },
   {
      name : "Tanishq",
      age : 21,
      branch : "CS",
      marks : 88,
      city : "Mumbai",
   },
   {
      name : "Sarthak",
```

```
    age : 20,
    branch : "IT",
    marks : 92,
    city : "Pune",
},
{
    name : "Anushree",
    age : 21,
    branch : "ENTC",
    marks : 87,
    city : "Nashik",
},
{
    name : "Rahul",
    age : 20,
    branch : "CS",
    marks : 76,
    city : "Mumbai",
},
{
    name : "Neha",
    age : 21,
```

```
    branch : "IT",
    marks : 97,
    city : "Pune",
  },
  {
    name : "Shreya",
    age : 20,
    branch : "ENTC",
    marks : 69,
    city : "Nashik",
  },
  {
    name : "sona",
    age : 21,
    branch : "IT",
    marks : 85,
    city : "Pune",
  },
]);
```

OUTPUT :

```
{
  acknowledged: true,
```

```
  insertedIds: {
    '0': ObjectId('6a8854382527ae40b6c1cd1e'),
    '1': ObjectId('6a8854382527ae40b6c1cd1f'),
    '2': ObjectId('6a8854382527ae40b6c1cd20'),
    '3': ObjectId('6a8854382527ae40b6c1cd21'),
    '4': ObjectId('6a8854382527ae40b6c1cd22'),
    '5': ObjectId('6a8854382527ae40b6c1cd23'),
    '6': ObjectId('6a8854382527ae40b6c1cd24'),
    '7': ObjectId('6a8854382527ae40b6c1cd25')
  }
}
```

Find All Students :

```
db.students.find();
```

output :

```
{
  _id: ObjectId('6a8854382527ae40b6c1cd1e'),
  name: 'sanskruti',
  age: NumberInt('20'),
  branch: 'IT',
  marks: NumberInt('98'),
  city: 'Nashik'
}
```

```
{
  _id: ObjectId('6a8854382527ae40b6c1cd1f'),
  name: 'Tanishq',
  age: NumberInt('21'),
  branch: 'CS',
  marks: NumberInt('88'),
  city: 'Mumbai'
}
{
  _id: ObjectId('6a8854382527ae40b6c1cd20'),
  name: 'Sarthak',
  age: NumberInt('20'),
  branch: 'IT',
  marks: NumberInt('92'),
  city: 'Pune'
}
{
  _id: ObjectId('6a8854382527ae40b6c1cd21'),
  name: 'Anushree',
  age: NumberInt('21'),
  branch: 'ENTC',
  marks: NumberInt('87'),
```

```
  city: 'Nashik'
}
{
  _id: ObjectId('6a8854382527ae40b6c1cd22'),
  name: 'Rahul',
  age: NumberInt('20'),
  branch: 'CS',
  marks: NumberInt('76'),
  city: 'Mumbai'
}
{
  _id: ObjectId('6a8854382527ae40b6c1cd23'),
  name: 'Neha',
  age: NumberInt('21'),
  branch: 'IT',
  marks: NumberInt('97'),
  city: 'Pune'
}
{
  _id: ObjectId('6a8854382527ae40b6c1cd24'),
  name: 'Shreya',
  age: NumberInt('20'),
```

branch: 'ENTC',

marks: NumberInt('69'),

city: 'Nashik'

}

{

_id: ObjectId('6a8854382527ae40b6c1cd25'),

name: 'sona',

age: NumberInt('21'),

branch: 'IT',

marks: NumberInt('85'),

city: 'Pune'

}

Find One Student :

db.students.findOne({name : "sanskruti"});

OUTPUT :

{

_id: ObjectId('6a8854382527ae40b6c1cd1e'),

name: 'sanskruti',

age: NumberInt('20'),

branch: 'IT',

marks: NumberInt('98'),

city: 'Nashik'

}

Find Students by Branch :

db.students.find({branch : "IT"});

OUTPUT :

All the students of branch IT will be displayed.

Find Students with Marks Not Equal to 85 :

db.students.find({ marks: { $ne: 85 }});

OUTPUT :

All the students whose marks except 85 will be displayed.

Find Students with Marks Greater Than 80 :

db.students.find({marks : {$gt : 80}});

OUTPUT :

All the students whose marks is greater than 80 will be displayed.

Find Students with Marks Less Than 80 :

db.students.find({marks : {$lt : 80}});

OUTPUT :

All the students whose marks is less than 80 will be displayed.

Find Students with Marks Greater Than or Equal to 80 :

db.students.find({ marks: { $gte: 80 } });

OUTPUT :

All the students whose marks is greater than or equal to 80 will be displayed.

Find Students with Marks Less Than or Equal to 80 :

```
db.students.find({ marks: { $lte: 80 } });
```

OUTPUT :

All the students whose marks is less than or equal to 80 will be displayed.

Find Students with Marks Between 80 and 90 :

```
db.students.find({marks : {$gte : 80,$lte : 90}});
```

OUTPUT :

All the students whose marks is between 80 and 90 will be displayed.

Find IT Students with Marks Greater Than 80 :

```
db.students.find({branch: "IT",marks: { $gt: 80 }});
```

OUTPUT :

All the students whose branch is IT and marks is greater than 80 will be displayed.

Find IT Students with Marks Greater Than 80 Using AND :

```
db.students.find({$and: [{ branch: "IT" },{ marks: { $gt: 80 } }]});
```

OUTPUT :

All the students whose branch = IT AND marks > 80 will be displayed. Both condition must be true.

Find Students from IT or CS :

```
db.students.find({$or: [{ branch: "IT" },{ branch: "CS" }]});
```

OUTPUT :

All the students whose branch = IT OR branch = CS will be displayed. Atleast one condition must be true.

Find Students from IT or CS Using IN :

db.students.find({branch: { $in: ["IT", "CS"] }});

OUTPUT :

All the students whose branch is either IT or CS will be displayed. It is used to match any values from a list. It is convenient when checking **one field against multiple values**.

Find Students Not from IT or CS :

db.students.find({branch: { $nin: ["IT", "CS"] }});

OUTPUT :

All the students who are not from IT OR CS will be displayed.

Display Selected Fields :

db.students.find({},{name : 1, marks : 1});

OUTPUT :

{

_id: ObjectId('6a8854382527ae40b6c1cd1e'),

name: 'sanskruti',

marks: NumberInt('98')

}

{

_id: ObjectId('6a8854382527ae40b6c1cd1f'),

name: 'Tanishq',

marks: NumberInt('88')

}

```
{
  _id: ObjectId('6a8854382527ae40b6c1cd20'),
  name: 'Sarthak',
  marks: NumberInt('92')
}
{
  _id: ObjectId('6a8854382527ae40b6c1cd21'),
  name: 'Anushree',
  marks: NumberInt('87')
}
{
  _id: ObjectId('6a8854382527ae40b6c1cd22'),
  name: 'Rahul',
  marks: NumberInt('76')
}
{
  _id: ObjectId('6a8854382527ae40b6c1cd23'),
  name: 'Neha',
  marks: NumberInt('97')
}
{
  _id: ObjectId('6a8854382527ae40b6c1cd24'),
```

name: 'Shreya',

marks: NumberInt('69')

}

{

_id: ObjectId('6a8854382527ae40b6c1cd25'),

name: 'sona',

marks: NumberInt('85')

}

Display Selected Fields Without ID :

db.students.find({},{_id : 0,marks :1, name :1});

OUTPUT :

Same output as above only id will not be displayed.

0 -> exclude

1 -> include

Sort Students by Marks Descending :

db.students.find().sort({marks : -1});

OUTPUT :

{

_id: ObjectId('6a8854382527ae40b6c1cd1e'),

name: 'sanskruti',

age: NumberInt('20'),

branch: 'IT',

```
  marks: NumberInt('98'),
  city: 'Nashik'
}
{
  _id: ObjectId('6a8854382527ae40b6c1cd23'),
  name: 'Neha',
  age: NumberInt('21'),
  branch: 'IT',
  marks: NumberInt('97'),
  city: 'Pune'
}
{
  _id: ObjectId('6a8854382527ae40b6c1cd20'),
  name: 'Sarthak',
  age: NumberInt('20'),
  branch: 'IT',
  marks: NumberInt('92'),
  city: 'Pune'
}
{
  _id: ObjectId('6a8854382527ae40b6c1cd1f'),
  name: 'Tanishq',
```

```
  age: NumberInt('21'),
  branch: 'CS',
  marks: NumberInt('88'),
  city: 'Mumbai'
}
{
  _id: ObjectId('6a8854382527ae40b6c1cd21'),
  name: 'Anushree',
  age: NumberInt('21'),
  branch: 'ENTC',
  marks: NumberInt('87'),
  city: 'Nashik'
}
{
  _id: ObjectId('6a8854382527ae40b6c1cd25'),
  name: 'sona',
  age: NumberInt('21'),
  branch: 'IT',
  marks: NumberInt('85'),
  city: 'Pune'
}
{
```

```
  _id: ObjectId('6a8854382527ae40b6c1cd22'),
  name: 'Rahul',
  age: NumberInt('20'),
  branch: 'CS',
  marks: NumberInt('76'),
  city: 'Mumbai'
}
{
  _id: ObjectId('6a8854382527ae40b6c1cd24'),
  name: 'Shreya',
  age: NumberInt('20'),
  branch: 'ENTC',
  marks: NumberInt('69'),
  city: 'Nashik'
}
```

Sort Students by Marks Ascending :

```
db.students.find().sort({marks : 1});
```

OUTPUT :

All the data will be displayed in ascending order of marks.

Find Topper :

```
db.students.find().sort({marks : -1}).limit(1);
```

OUTPUT :

{

_id: ObjectId('6a8854382527ae40b6c1cd1e'),

name: 'sanskruti',

age: NumberInt('20'),

branch: 'IT',

marks: NumberInt('98'),

city: 'Nashik'

}

Find Top 3 Students :

db.students.find().sort({marks : -1}).limit(3);

OUTPUT :

{

_id: ObjectId('6a8854382527ae40b6c1cd1e'),

name: 'sanskruti',

age: NumberInt('20'),

branch: 'IT',

marks: NumberInt('98'),

city: 'Nashik'

}

{

_id: ObjectId('6a8854382527ae40b6c1cd23'),

name: 'Neha',

```
  age: NumberInt('21'),
  branch: 'IT',
  marks: NumberInt('97'),
  city: 'Pune'
}
{
  _id: ObjectId('6a8854382527ae40b6c1cd20'),
  name: 'Sarthak',
  age: NumberInt('20'),
  branch: 'IT',
  marks: NumberInt('92'),
  city: 'Pune'
}
```

Find Lowest 3 Students :

```
db.students.find().sort({marks : -1}).skip(1).limit(3);
```

OUTPUT :

```
{
  _id: ObjectId('6a8854382527ae40b6c1cd24'),
  name: 'Shreya',
  age: NumberInt('20'),
  branch: 'ENTC',
  marks: NumberInt('69'),
```

```
  city: 'Nashik'
}
{
  _id: ObjectId('6a8854382527ae40b6c1cd22'),
  name: 'Rahul',
  age: NumberInt('20'),
  branch: 'CS',
  marks: NumberInt('76'),
  city: 'Mumbai'
}
{
  _id: ObjectId('6a8854382527ae40b6c1cd25'),
  name: 'sona',
  age: NumberInt('21'),
  branch: 'IT',
  marks: NumberInt('85'),
 :
}
```

Count IT Students :

```
db.students.countDocuments({branch: "IT"});
```

OUTPUT : 4

Update Student City :

```
db.students.updateOne({name : "sanskruti"},{$set : {city : "Pune"}});
```

OUTPUT :

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: NumberInt('1'),
  modifiedCount: NumberInt('1'),
  upsertedCount: NumberInt('0')
}
```

Update Multiple Student Fields :

```
db.students.updateOne({ name: "Sanskruti" },{ $set: { age: 21,city:
"Nashik",marks: 95 }});
```

OUTPUT :

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: NumberInt('0'),
  modifiedCount: NumberInt('0'),
  upsertedCount: NumberInt('0')
}
```

```
db.students.updateOne({ name: "Sanskruti" },{ $inc: { marks: 5 } });
```

OUTPUT :

{

acknowledged: true,

insertedId: null,

matchedCount: NumberInt('0'),

modifiedCount: NumberInt('0'),

upsertedCount: NumberInt('0')

}

Increase Student Marks Using Increment :

db.students.updateMany({ branch: "IT" },{ $inc: { marks: 2 } });

OUTPUT :

{

acknowledged: true,

insertedId: null,

matchedCount: NumberInt('4'),

modifiedCount: NumberInt('4'),

upsertedCount: NumberInt('0')

}

Replace a Student Record :

db.students.replaceOne({ name: "Rahul" },{ name: "Rahul",age: 21,branch: "IT",marks: 90,city: "Nashik"}

);

OUTPUT :

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: NumberInt('1'),
  modifiedCount: NumberInt('1'),
  upsertedCount: NumberInt('0')
}
```

It replace the entire document.

Delete One Student :

```
db.students.deleteOne({name : "Rahul"});
```

OUTPUT :

```
{
  acknowledged: true,
  deletedCount: NumberInt('1')
}
```

Delete Student Based on Marks

```
db.students.deleteOne({marks :{$lt : 70}});
```

OUTPUT :

```
{
  acknowledged: true,
  deletedCount: NumberInt('1')
}
```

Delete Students by Branch :

```
db.students.deleteMany({branch : "ENTC"});
```

OUTPUT :

```
{
  acknowledged: true,
  deletedCount: NumberInt('1')
}
```

Delete Students Based on Marks :

```
db.students.deleteMany({marks : {$lt : 60}});
```

OUTPUT :

```
{
  acknowledged: true,
  deletedCount: NumberInt('0')
}
```

Find Distinct Branches :

```
db.students.distinct("branch");
```

OUTPUT :

```
[
  'CS',
  'IT'
]
```

Calculate Average Marks :

```
db.students.aggregate([
  {
    $group: {
      _id: null,
      averageMarks: { $avg: "$marks" }
    }
  }
]);
```

OUTPUT :

```
{
 _id: null,
 averageMarks: Double('93.6')
}
```

Find Highest Marks :

```
db.students.aggregate([
  {
    $group: {
      _id: null,
      highestMarks: { $max: "$marks" }
    }
  }
]);
```

OUTPUT :

```
{
  _id: null,
  highestMarks: NumberInt('100')
}
```

Find Lowest Marks :

```
db.students.aggregate([
   {
      $group: {
         _id: null,
         lowestMarks: { $min: "$marks" }
      }
   }
]);
```

OUTPUT :

```
{
  _id: null,
  lowestMarks: NumberInt('87')
}
```

Count Students Branch-wise :

```
db.students.aggregate([
   {
```

```
    $group: {
      _id: "$branch",
      count: { $sum: 1 }
    }
  }
]);
```

OUTPUT :

```
{
 _id: 'IT',
 count: NumberInt('4')
}
{
 _id: 'CS',
 count: NumberInt('1')
}
```

Calculate Average Marks Branch-wise :

```
db.students.aggregate([
  {
    $group: {
      _id: "$branch",
      averageMarks: { $avg: "$marks" }
    }
```

```
  }
]);
```

OUTPUT :

```
{
 _id: 'CS',
 averageMarks: NumberInt('88')
}
{
 _id: 'IT',
 averageMarks: NumberInt('95')
}
```

Find Branch with Highest Average Marks :

```
db.students.aggregate([
  {
    $group: {
      _id: "$branch",
      averageMarks: { $avg: "$marks" }
    }
  },
  {
    $sort: { averageMarks: -1 }
  },
```

```
    {
        $limit: 1
    }
]);
```

OUTPUT :

```
{
  _id: 'IT',
  averageMarks: NumberInt('95')
}
```